

FICTIONAL DEMO SUBMISSION

For LMS grading-workflow demonstration only

Assignment 2 - Polynomial and Rational Functions

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Selected responses

These original responses are intentionally concise and are provided only to demonstrate Moodle submission, download, and grading workflows.

Question 1

P has degree 4 and leading coefficient -1. The zeros are -2 (multiplicity 1), 1 (multiplicity 2), and 4 (multiplicity 1). Both ends fall. $P(0) = -8$, so the y-intercept is (0,-8).

Question 4

$R(x)$ factors to $(x - 3)(x + 2)/[(x - 2)(x + 2)]$. The original restrictions are $x \neq -2$ and $x \neq 2$. There is a hole at $(-2, 5/4)$, a vertical asymptote at $x = 2$, a horizontal asymptote at $y = 1$, x-intercept (3,0), and y-intercept (0,3/2).

Question 5

Restrictions: $x \neq 1, -2$. Cross-multiplying gives $2(x + 2) = 3(x - 1)$, so $2x + 4 = 3x - 3$ and $x = 7$. Substitution gives $2/6 = 3/9 = 1/3$, so it is valid.
